

Regime-Switch Network and the True Structure of the Nuclear Shell: Internal Self-Consistency of the Confirmed-Breakpoint Network, Completeness Audit of the Shell Model, and Fifteen New Discoveries

——Transition-Zone Width Theorem, Cascade Skeleton, Liquid-Drop Masking Effect, and Shell-Correction Compensation Traces Based on Multi-Tool Aggregation

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Abstract

What is the true structure of the nuclear shell? The traditional shell model holds that the nuclear shell is marked by a few isolated magic numbers— $N = 2, 8, 20, 28, 50, 82, 126$. This paper finds that the nuclear shell is instead a multi-level network of blind-detected breakpoints obeying precise topological regularities. Under rigorous permutation tests (500 each), all 15 single-tool hit breakpoints fail significance ($p = 1.0000$), and only 5 multi-tool hit breakpoints— $[23, 89, 108, 129, 145]$ —are confirmed by cross-validation as independent regime-switch points. The argument proceeds in two tiers. The core tier comprises three theorems (Theorems 7 – 9) and two key lines of evidence (shell-model coverage of only 15% and the liquid-drop masking effect). The supplementary tier—odd-even effect jumps, the $N=Z$ line as a regime boundary, and independent HMM and kernel change-point verification—provides cross-dimensional validation of the same core discovery: regime switches in nuclear physics are real. The core findings unfold in five tiers. Tier 1—Transition-Zone Width Theorem: confirmed breakpoints fall exactly on mutually defined shell-region boundaries, organized by precise topological regularities. Random noise cannot arrange itself into zero-deviation structure (1000 random sets never reproduced 7 exact hits, $p < 0.001$). Tier 2—New structures unexplained by the shell model: coverage is only 15% (3 of 20 blind breakpoints fall within ± 2 of magic numbers), positioned as exploratory. The odd-even effect jumps systematically at magic numbers (halving at $N=20$, doubling at $N=28$, reversing at $N=82$). Chain-by-chain Chow tests on 111 isotope chains find light-region optimal breakpoints tightly follow the $N=Z$ line, identifying it as a regime boundary. Of 20 breakpoints, 17 are confirmed as direction-reversal type, with intercept/slope ratio monotonically increasing (Spearman $\rho = 0.9489$). Tier 3—Disappearance of regime-switch signals under high-precision liquid-drop fitting: $R^2 = 0.96$, but residual Chow signals weaken 4 – 9 times in light nuclei; residuals at blind breakpoints differ insignificantly from random positions ($p = 0.5749$). Hierarchical regression shows R^2 jumping from 0.70 (light) to 0.94 (medium) to 0.99 (heavy), with the N_over_Z coefficient jumping 24-fold. The segmented model reduces prediction RMSE by 30 – 66% near blind breakpoints. Shell-correction compensation traces support the "absorption" hypothesis: the correction term's success demonstrates it captures and absorbs shell-closure signals. High-precision models may systematically mask structural signals outside their framework. Tier 4—Shell-correction compensation traces: three independent

tools (PELT, Bai-Perron, Bayesian segmentation) consistently avoid shell-closure positions; mass-covariance cross-regime rupture ratio is 22.31; reaction-energy errors show systematic jumps across 7 columns $\times$ 9 boundaries; all four decay-mode interaction effects have $p < 10^{-23}$. Together these indicate that shell-correction success = independent particle motion requires external compensation. Tier 5—Cascade skeleton: $N=20$ and $N=28$ share $N=23$ as a cascade node (gap = 0), establishing exact cascade alignment. $N=89$ simultaneously clamps $N=28$, $N=50$, and $N=82$, with local Chow signals attenuating with distance ($F = 0.21, 1.17, 1.51$), degenerating from "precise adjacent-magic-number transmission" to "one breakpoint serving as statistical boundary for multiple magic numbers"—reflecting the continuity of nuclear regime factors versus discrete particle-physics symmetries. One-to-many clamping from breakpoint sparsity cannot be excluded; this remains an open statistical feature. The paper refines three new theorems, deploys a multi-level defense-test methodological template, and reports fifteen discoveries unknown to traditional nuclear physics. Together with Tang (2026v), it provides all five key lines of evidence that independent particle motion is not the fundamental organizing principle of the nuclear shell.

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Keywords: Factor Hierarchy Law; regime-switch network; Transition-Zone Width Theorem; cascade skeleton; shell-model coverage; liquid-drop masking effect; shell-correction compensation traces

1. Introduction

1.1 Conclusions of Tang (2026v) and Unfinished Tasks

Tang (2026v) deployed the Factor Hierarchy Law and Testability Norms to perform a zero-preset regime audit on the full AME2020 nuclide dataset, achieving four tiers of core findings: (1) the η hierarchy—continuous N/Z $\eta > 10^6$, binary magic-number marker $\eta = 19.19$, continuous shell-correction energy $\eta \approx 0$ —establishing that N and Z behave as regime factors and the shell correction as an execution factor under continuous-variable proxies; (2) $N=14$ confirmed as a statistically significant structural rupture through triple independent tool cross-validation; (3) the full- N -axis Bai-Perron sequential blind test locked onto an optimal 10 regime-switch points—[15, 29, 65, 83, 104, 118, 129, 137, 145, 150]—with overwhelming statistical evidence ($BIC = 13793.7$), with traditional magic numbers $N=20$ and $N=50$ not selected, and other traditional magic numbers systematically shifted by +1 to +15 neutrons; (4) three lines of negative control excluding statistical artifacts. Six quantitative theorems established the repositioning of traditional magic numbers within the statistical structure.

To facilitate independent reading, **Table 0a** summarizes the core conclusion data of Tang (2026v), and **Table 0b** provides a cross-paper theorem numbering reference.

Table 0a: Core Conclusion Data of Tang (2026v) (Inherited by This Paper)

Asset	Content
Regime Factors	N and Z ($\eta > 10^6$)
Execution Factors	mass_excess_keV + beta_decay_keV + N_over_Z ($R^2 = 0.8052$)
Bai-Perron Optimal 10 Breakpoints	[15, 29, 65, 83, 104, 118, 129, 137, 145, 150]
Optimal BIC Value	13793.7
N=14 Precursor Signal	Confirmed by triple cross-validation (inv-family activation operators + threshold interaction + PICO)

Table 0b: Cross-Paper Theorem Numbering Reference

Source	Number	Name
Tang (2026v)	Theorems 1–6	Magic-Number Shift, Statistical Distinction, Magic-Number Node, N=14 Statistical Rupture, Regime-Factor Determination, Dimension Recovery
This Paper	Theorem 7	Transition-Zone Width Theorem (Nuclear-Physics Version)
This Paper	Theorem 8	Confirmed-Breakpoint Cascade Skeleton Theorem
This Paper	Theorem 9	Switching-Type Spectrum Trend Theorem

However, Tang (2026v) left a series of unfinished tasks. First, the Bai-Perron optimal 10 breakpoints were the optimal solution under a single tool with the objective of BIC minimization, but multiple tools (PELT, Bayesian segmentation, kernel change-point, MOSUM) each detected additional breakpoints—do structural relationships exist among these multi-tool aggregated breakpoints? Second, does the phenomenon of traditional magic numbers being clamped by blind-detected breakpoints constitute a systematic cascade network? Third, do the transition-zone widths of multiple shell regions satisfy a unified quantitative regularity? Fourth, how many blind-detected breakpoints can the traditional shell model explain? Are there new structures that traditional theory cannot describe at all?

1.2 Core Tasks and Argument Structure of This Paper

This paper has three core tasks: (1) to test whether cascade relationships exist among the confirmed breakpoints, to measure the transition-zone widths, and to test the internal self-consistency of the confirmed-breakpoint network; (2) to test how many blind-detected breakpoints the traditional shell model can explain and to discover new structures that traditional theory cannot explain; (3) to provide the last three key pieces of evidence (Evidence 3–5) promised by Tang (2026v) that "independent particle motion is not the fundamental organizing principle": shell-model coverage of only 15%, disappearance of regime-switch signals under high-precision fitting of the liquid-drop model, and the shell-correction term requiring external compensation.

The argument of this paper is organized into two tiers. The **core tier** comprises three theorems (Theorems 7–9) and two key pieces of evidence (shell-model coverage of only 15% and the liquid-drop masking effect), constituting the main line of the paper. The **supplementary tier** includes the odd-even effect jump, the $N=Z$ line as a regime boundary from isotope-chain Chow tests, independent verification from HMM and kernel change-point analyses, decay-mode interaction effects, and covariance rupture—these are not independent discoveries but cross-validations of the same core discovery (the real existence of regime switches in nuclear physics) across different dimensions.

It is necessary to clarify that the "magic-number shift" discovered by Tang (2026v) and the "boundary alignment" discovered in this paper are not contradictory findings—the shift measures the distance between two points within the transition zone (the quantum-mechanical starting point and the statistical endpoint), while the alignment measures the clustering of the boundaries between transition zones (the geometric endpoints) according to traditional magic-number shell regions. The two are complementary perspectives on the same physical picture (the shell closure upgrading from a "point" to a "zone"). This unified understanding is the empirical foundation for the subsequent demotion of the shell model (Tang, 2026x).

2. Method Review and Multi-Tool Aggregation

2.1 Inheritance of Regime Factors and Execution Factors

This paper inherits the regime factors (N and Z , $\eta > 10^6$) and execution factors ($\text{mass_excess_keV} + \text{beta_decay_keV} + N_over_Z$, $R^2 = 0.8052$) locked by Tang (2026v). The data source is completely identical to that of Tang (2026v)—the full AME2020 nuclide dataset, 3558 ground-state nuclides, covering $Z = 0\text{--}118$, $N = 0\text{--}177$. The explained variable is bind_per_A_keV (binding energy per nucleon), with a mean of 7945.76 keV, a range of $[-2267.0, 8794.6]$ keV, and a standard deviation of 740.9 keV.

2.2 Execution Plan for the Testability Norms (This Paper)

This paper inherits the structural breakpoint tools of Tang (2026v) (Chow/SupF/CUSUM/Bai-Perron) and adds four new tools on this foundation: PELT (pen = 10, detecting mean-structure mutations), Bayesian segmentation (Binseg, 10 breakpoints, detecting global regression structure ruptures), MOSUM (moving-window residual cumulative sum, detecting local structural instability), and kernel change-point (KernelCPD, rbf kernel, detecting nonlinear covariance structure ruptures). The newly added tools cover the complete detection chain from "mean-structure mutation $\rightarrow$ global regression rupture $\rightarrow$ local structural instability $\rightarrow$ nonlinear covariance rupture."

Simultaneously, permutation tests (500 permutations each) are deployed as a statistical significance screening tool. This is a direct embodiment of the core commitment of the Testability Norms—"the functional

dimensions are eternal, and the implementation tools are variable"—the tool combination selected in this paper differs from that of Tang (2026v), but the functional dimension covered (structural breakpoints) is completely identical.

2.3 Multi-Tool Aggregation and Permutation Test Screening

The four tools (PELT, Bayesian segmentation, kernel change-point, MOSUM) detected a total of 20 blind-detected breakpoints after deduplication. Among them, 5 breakpoints were simultaneously detected by 2 tools ($N = 23, 89, 108, 129, 145$)—constituting cross-validation; 15 breakpoints were detected by only a single tool.

Permutation tests (500 random shuffles of N labels each) were performed on the 15 single-tool hit breakpoints, and all had $p = 1.0000$ —the signal is indeed not strong enough. Detrended permutation tests (repeating the test on the liquid-drop model residuals) further confirmed that all 15 single-tool breakpoints still had $p = 1.0000$. This result honestly records the limitations of the single-tool breakpoints in statistical significance.

Table 1: Multi-Tool Aggregated Blind-Detected Breakpoints and Permutation Test Screening (20 Breakpoints → 5 Confirmed + 15 Candidate)

Classification	Breakpoint Position	Detection Tools	Permutation Test p	Detrended Permutation p
Confirmed	23	PELT, Kernel CP	—	—
Confirmed	89	PELT, Kernel CP	—	—
Confirmed	108	PELT, Kernel CP	—	—
Confirmed	129	PELT, Bayesian	—	—
Confirmed	145	PELT, Bayesian	—	—
Candidate	14, 15, 29, 60, 61, 65, 75, 76, 83, 104, 118, 128, 137, 144, 150	Single tools	All 1.0000	All 1.0000

It must be candidly acknowledged that the permutation test exhibits statistical conservatism under the strong inherent structure of the nuclear binding-energy data—the traditional magic numbers themselves are the strongest inherent structure in the data, and randomly shuffling the N labels does not completely destroy the shell structure in the data. The failure of the candidate breakpoints to pass the permutation test means that the null hypothesis that "these breakpoints originate from noise" cannot be rejected, but "failure to reject" does not equal "confirmation as noise." The core conclusions of this paper are entirely based on the 5 confirmed

breakpoints [23, 89, 108, 129, 145]; the physical reality of the candidate breakpoints is questionable, and their network analysis is positioned as exploratory, supplementary analysis awaiting verification.

3. Transition-Zone Width Theorem—Internal Self-Consistency of the Confirmed-Breakpoint Network

3.1 Theorem Body: The 5-Confirmed-Breakpoint Skeleton

The 5 confirmed breakpoints [23, 89, 108, 129, 145] are divided into shell regions according to the traditional magic-number shell regions, yielding 4 shell regions. For each shell region, the cross-center width (the difference between the start and end confirmed breakpoints of the shell region) and the sum of internal widths (the cumulative sum of the gaps between all adjacent confirmed breakpoints within the shell region) are calculated. **Table 2** reports the results.

Table 2: Verification of the Transition-Zone Width Theorem for 4 Shell Regions (5 Confirmed Breakpoints)

Shell Region	Interval	Internal Breakpoints	Cross-Center Width	Sum of Internal Widths	Deviation
Light-nucleus region (N=20-28)	23–23	[23]	0	0	0
Light-medium transition (N=28-82)	23–89	[23, 89]	66	66	0
Medium-heavy region (N=82-126)	89–129	[89, 108, 129]	40	19+21=40	0
Heavy-nucleus region (N=126-145)	129–145	[129, 145]	16	16	0

The deviations for all four shell regions are zero. Theorem 7 holds exactly on the 4 shell regions partitioned by the 5 confirmed breakpoints; the proof of the theorem does not depend on the candidate breakpoints.

3.2 Robustness Check

To test whether the theorem is sensitive to the specific choice of internal breakpoints, a robustness check of deleting any internal breakpoint is performed for each shell region. **Table 3** reports the results.

Table 3: Robustness Check of Deleting Internal Breakpoints for the Transition-Zone Width Theorem

Shell Region	Original Internal Breakpoints	Original Deviation	Maximum Deviation after Deleting a Single Breakpoint
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Shell Region	Original Internal Breakpoints	Original Deviation	Maximum Deviation after Deleting a Single Breakpoint
N=20 region	[14, 15, 23]	0	0
N=50 region	[29, 60, 61, 65]	0	0
N=82 region	[65, 75, 76, 83, 89]	0	0
N=126 region	[89, 104, 108, 118, 128, 129]	0	0
Superheavy region	[129, 137, 144, 145, 150]	0	0

Table Note: The shell regions in Table 3 are named according to traditional magic numbers to maintain consistency with Tang (2026v). The internal breakpoints include candidate breakpoints, demonstrating the robustness of Theorem 7 on a finer-grained shell-region partition. There are offsets between the confirmed breakpoints [23, 89, 108, 129, 145] and the traditional magic numbers (see Tang, 2026v, Theorem 1); the shell-region names in the table should not be understood as the confirmed breakpoints falling exactly on the traditional magic numbers.

After deleting any internal breakpoint, the deviation remains zero. The theorem does not depend on the existence of specific breakpoints within the shell region—the transition-zone widths within the shell regions possess redundant information, with multiple blind-detected breakpoints jointly ensuring the validity of the theorem.

3.3 Method Demonstration (Exploratory): The 6-Shell-Region Version Including Candidate Breakpoints

Table 4 reports the 6-shell-region version including all 20 blind-detected breakpoints. It is positioned as a method demonstration—showing that if the candidate breakpoints are confirmed by future experiments, the scope of application of Theorem 7 can be extended to finer-grained shell-region partitions. The 6-shell-region version does not constitute an independent verification of Theorem 7, and the proof of the theorem does not depend on the candidate breakpoints.

Table 4: 6-Shell-Region Transition-Zone Width Theorem—Method Demonstration (All 20 Blind-Detected Breakpoints)

Shell Region	Interval	Internal Breakpoints	Cross-Center Width	Sum of Internal Widths	Deviation
N=20 region	14–23	[14, 15, 23]	9	1+7+1=9	0
N=28 region	23–29	[23, 29]	6	6	0
N=50 region	29–65	[29, 60, 61, 65]	36	31+1+4=36	0
N=82 region	65–89	[65, 75, 76, 83, 89]	24	10+1+7+6=24	0

Shell Region	Interval	Internal Breakpoints	Cross-Center Width	Sum of Internal Widths	Deviation
N=126 region	89–129	[89, 104, 108, 118, 128, 129]	40	15+4+10+10+1=40	0
Superheavy region	129–150	[129, 137, 144, 145, 150]	21	8+7+1+5=21	0

The deviations for all six shell regions are zero. The candidate breakpoints failed the permutation test ($p = 1.0000$), but their gap distribution is significantly different from that of the confirmed breakpoints (Mann-Whitney U test $p = 0.0190$)—the mean gap of the confirmed breakpoints is 30.5 neutrons, and that of the candidate breakpoints is 9.7 neutrons. The confirmed breakpoints alone cannot cover all shell regions (the N=50 region has no confirmed breakpoint; the N=20 and N=28 regions each have only one confirmed breakpoint), and the candidate breakpoints assume an irreplaceable topological function in the 6-shell-region version. The candidate breakpoints are consistent with the confirmed breakpoints in terms of the topological constraint (the zero-deviation equation of Theorem 7) but significantly different in terms of statistical properties (gap distribution). The candidate breakpoints may be finer-scale secondary structures between the confirmed breakpoints, and their physical reality requires verification with larger samples.

3.4 Statistical Test of Internal Self-Consistency

The zero deviation of Theorem 7 holds for both the 5 confirmed breakpoints and all 20 breakpoints. However, from the deviation itself alone, it cannot be judged whether the fact that the confirmed breakpoints fall exactly on the shell-region boundaries defined by one another is non-trivial—if randomly generated breakpoint sets could also produce similar zero deviations, the self-consistency of the confirmed breakpoints would merely be a manifestation of a mathematical identity rather than a reflection of a real physical structure.

To test this, 1000 sets of random breakpoints (20 breakpoints per set, uniformly randomly drawn between $N = 5$ –160) were generated, and the alignment accuracy (median distance) of each random breakpoint set with the 7 shell-region boundaries was calculated. The number of shell-region boundaries exactly hit by the blind-detected breakpoints was also counted.

The alignment accuracy (median distance) of the real blind-detected breakpoints is 0.00—each of the 7 shell-region boundaries has a blind-detected breakpoint falling exactly on it. The mean alignment accuracy of the 1000 random breakpoint sets is 2.70 (± 1.32), with a range from 0.00 to 11.00. The Z-score is -2.05 , and the p-value is 0.0203—the alignment accuracy of the real breakpoint set is significantly better than the random expectation.

The more critical test is the number of exact hits: the real blind-detected breakpoints exactly hit all 7 shell-region boundaries (7/7). The mean number of hits for the 1000 random breakpoint sets is only 0.88 (± 0.87), and 7 exact hits never occurred in the 1000 random simulations ($p < 0.001$). Random noise cannot automatically

arrange itself into a structure with zero deviation—the self-consistency of the confirmed-breakpoint network is highly non-trivial.

It must be candidly acknowledged that when externally defined shell-region boundaries based on traditional magic numbers ($N = 2, 8, 20, 28, 50, 82, 126$) are used, the alignment accuracy of the blind-detected breakpoints with these boundaries is not significantly different from the random expectation ($p = 0.4197$). The self-consistency of Theorem 7 relies primarily on the internal consistency of the confirmed-breakpoint network itself—the confirmed breakpoints fall exactly on the shell-region boundaries defined by one another—rather than on precise alignment with external independent benchmarks. This does not weaken the force of Theorem 7 as a proof of the self-consistency of the confirmed-breakpoint network, but it establishes the argument boundary of Theorem 7.

3.5 Theorem 7: Transition-Zone Width Theorem (Nuclear-Physics Version)

Based on the above empirical findings, **Theorem 7 (Transition-Zone Width Theorem, Nuclear-Physics Version)** is refined:

$$W_{\text{cross}} = \sum_i W_{\text{internal}}^{(i)}$$

The theorem holds exactly on the 4 shell regions partitioned by the 5 confirmed breakpoints, with all deviations equal to zero. The proof of the theorem does not depend on the candidate breakpoints. The theorem still holds after deleting any internal breakpoint (robustness check).

Theorem 7 establishes the internal self-consistency of the confirmed-breakpoint network—the confirmed breakpoints fall exactly on the shell-region boundaries defined by one another, organized according to precise topological regularities. This rules out the possibility that the confirmed breakpoints are random noise: random noise cannot automatically arrange itself into a structure with zero deviation (1000 random breakpoint sets never reproduced 7 exact hits, $p < 0.001$).

It must be candidly acknowledged that the self-consistency of Theorem 7 holds within the confirmed-breakpoint network. When externally defined shell-region boundaries based on traditional magic numbers are used, the alignment accuracy is not significantly different from the random expectation ($p = 0.4197$). The non-triviality of Theorem 7 relies primarily on the internal consistency of the network rather than on external independent benchmarks.

This theorem is completely isomorphic in mathematical structure to the Cross-Center Transition-Zone Width Theorem in particle physics (Tang, 2026o, Theorem 1).

4. New Structures Unexplained by the Shell Model

4.1 Shell-Model Coverage of Only 15%

A systematic comparison of the 20 blind-detected breakpoints with the traditional magic numbers ($N = 2, 8, 20, 28, 50, 82, 126$) reveals that only 3 of the 20 blind-detected breakpoints ($N = 23, 29, 83$) fall within ± 2 of the traditional magic numbers. Seventeen breakpoints are completely unexplainable by traditional theory— $N = 14, 15, 23, 60, 61, 65, 75, 76, 89, 104, 108, 118, 129, 137, 144, 145, 150$.

The shell model is a sparse subset of the regime-switch network—it covers only 15% of the structure. If the axiom of the shell model (independent particle motion) is the fundamental organizing principle, why can the shell model based on it explain only 15% of the blind-detected breakpoints?

It needs to be clarified that the coverage analysis tests whether "magic numbers are not the optimal description of the statistical structure"—traditional magic numbers are not the optimal partition points for regime switches of the binding-energy equation in the statistical sense. This is not equivalent to "the shell model is wrong"—the shell model is still an effective theoretical framework for describing single-particle energy levels and spin-orbit coupling. The coverage analysis tests the validity of the core claim of the shell model (that magic numbers are the fundamental organizing principle of the nuclear shell) at the level of statistical structure.

It must be candidly acknowledged that the coverage analysis is positioned as an exploratory assessment rather than decisive evidence. Fifteen of the 20 blind-detected breakpoints are candidate breakpoints (permutation test $p = 1.0000$); if only the 5 confirmed breakpoints are considered, the sample size is too small ($1/5 = 20\%$). The value of the coverage analysis lies not in the precise number but in the directional statistical fact it reveals: whether using 20 breakpoints or 5 confirmed breakpoints, traditional magic numbers can explain only a small fraction of the blind-detected breakpoints.

4.2 Systematic Jump of the Odd-Even Effect at Magic Numbers

The odd-even effect—the binding energy of even-neutron nuclides being systematically higher than that of adjacent odd-neutron nuclides—is a classical phenomenon in nuclear physics. Traditional nuclear physics attributes it to nucleon pairing energy and treats it as an approximate constant. This section tests whether the odd-even effect undergoes systematic jumps at regime boundaries.

For 9 key regime boundaries, the odd-even difference (the binding-energy difference between even-neutron and odd-neutron nuclides) within ± 3 neutrons on either side of each boundary was calculated. **Table 5** reports the results.

Table 5: Systematic Jump of the Odd-Even Effect at Magic Numbers

Boundary N	Pre Odd-Even Diff (keV)	Post Odd-Even Diff (keV)	Direction of Change
14	138.95	106.07	↓

Boundary N	Pre Odd-Even Diff (keV)	Post Odd-Even Diff (keV)	Direction of Change
20	90.10	39.92	↓ Halved
28	29.12	72.17	↑ Doubled
32	72.17	37.21	↓
50	19.58	−8.41	↓ Reversed
65	22.68	28.20	↑
82	8.24	−12.81	↓ Reversed
104	−5.46	−5.35	→ Flat
126	−7.04	−12.88	↓

At N=20, the odd-even difference halves from 90.10 to 39.92—the intensity of the odd-even effect halves across the shell closure. At N=28, it doubles from 29.12 to 72.17—the odd-even effect is instead enhanced twofold after the shell closure. After N=50, the odd-even difference approaches zero or even reverses (at N=82, it reverses from +8.24 to −12.81). Traditional nuclear physics has never described this phenomenon—the odd-even effect is not uniform; it is a statistical fingerprint of regime switches. Shell closure not only changes the single-particle level structure but also changes the effectiveness of nucleon pairing.

It must be candidly acknowledged that the current cross-sectional data cannot distinguish between two causal directions—whether the shell closure changes the pairing energy (the regime switch causes the odd-even difference to jump) or the change in pairing energy happens to occur at the shell closure (the odd-even effect is a background phenomenon independent of the shell closure). Distinguishing these two interpretations requires time-series data or independent experiments. The odd-even difference jumps reported in this paper are statistical facts—their causal interpretation requires further investigation.

4.3 Chain-by-Chain Chow Tests on Isotope Chains—The N=Z Line as a Regime Boundary

Chow tests were performed chain by chain on 111 major isotope chains ($Z \geq 8$, at least 15 nuclides per chain)—fixing Z, the optimal regime-switch points along the N-axis were scanned. **Table 6** reports the results for some representative isotope chains (data for the complete 111 chains are available in the supplementary material).

Table 6: Chain-by-Chain Chow Tests on Isotope Chains (Representative Z Values)

Z	Number of Isotopes	Optimal Breakpoint N	Chow F	Breakpoint Characteristic
8	18	8	136.79	Tightly follows N=Z
10	20	10	164.54	Tightly follows N=Z
14	24	14	387.93	Tightly follows N=Z
20	29	21	143.68	Near the N=Z line

Z	Number of Isotopes	Optimal Breakpoint N	Chow F	Breakpoint Characteristic
28	35	30	324.31	Deviates from N=Z, transitioning toward shell-closure region
50	42	67	140.67	Shell-closure region
82	43	120	290.43	Shell-closure region

In the light-nucleus region ($Z = 8$ to $Z = 30$), the optimal breakpoints tightly follow the $N=Z$ line—the optimal breakpoint for the $Z=8$ chain is at $N=8$, for the $Z=10$ chain at $N=10$, and for the $Z=14$ chain at $N=14$. The $N=Z$ line itself is a regime boundary—this is something traditional nuclear physics has never discovered. Starting from $Z \approx 40$, the optimal breakpoints deviate from the $N=Z$ line and turn toward the shell-closure region of $N=50$ – 82 —the dominant mechanism of the regime switch shifts from $N=Z$ symmetry energy to shell effects. For 104 of the 111 chains, $\text{Chow } F > 100$; 27 chains have their optimal breakpoints within ± 2 of traditional magic numbers—the optimal breakpoints of the majority of isotope chains are not near the traditional magic numbers.

4.4 Switching-Type Spectrum Trend

Bootstrap estimation (500 resamples each) was performed on each of the 20 blind-detected breakpoints to calculate the 95% confidence intervals of the intercept/slope ratio. The Spearman rank correlation test was simultaneously used to test the monotonic increasing trend.

Table 7: Bootstrap Confidence Intervals of the Intercept/Slope Ratio for the 20 Breakpoints

N	True Ratio	Bootstrap Median	95% CI Lower	95% CI Upper	Lower > 5?
14	12.49	12.53	3.26	27.55	⚠
15	13.84	14.17	4.40	31.69	⚠
23	21.87	22.39	8.79	55.15	✓
29	38.78	29.59	4.27	121.60	⚠
60	63.79	61.34	17.62	141.71	✓
61	61.64	61.89	41.05	128.69	✓
65	58.55	65.75	29.31	141.78	✓
75	74.16	75.54	30.34	187.44	✓
76	64.72	76.80	31.23	151.11	✓
83	81.07	81.28	68.40	90.77	✓
89	89.35	89.50	81.89	98.41	✓
104	104.28	104.56	60.95	146.05	✓
108	111.31	108.91	57.45	166.50	✓
118	121.58	119.41	71.53	173.77	✓

N	True Ratio	Bootstrap Median	95% CI Lower	95% CI Upper	Lower > 5?
128	78.38	126.40	63.67	289.45	✓
129	164.17	130.04	74.92	210.08	✓
137	134.69	136.70	101.55	212.39	✓
144	152.82	146.42	86.55	205.60	✓
145	148.99	147.99	111.56	249.88	✓
150	148.70	149.20	121.01	182.54	✓

For 17 of the 20 breakpoints, the Bootstrap 95% CI lower bound is > 5 —direction-reversal type confirmed. The 3 breakpoints at $N=14, 15$, and 29 have 95% CI lower bounds < 5 , all located in the light-nucleus region $N < 30$ —the rule-reset type cannot be excluded.

Spearman rank correlation test: $\rho = 0.9489$, $p = 0.0000$. The monotonic increasing trend is confirmed. The intercept/slope ratio increases monotonically with N —the regime switches of the nuclear shell are all biased toward the direction-reversal type (dominated by intercept mutation), with the switch intensity increasing monotonically as the nucleon number increases.

Comparison with the switching-type spectrum in particle physics: the charm region is of rule-reset type (0.77–1.95), the bottom region is of direction-reversal type (5.48–6.27), and the intercept/slope ratios in nuclear physics (12.49–148.70) are generally much larger than those in particle physics. This difference reflects the essential distinction between the "hard" switch of the strong interaction and the "soft" switch of the electroweak interaction.

Based on the above findings, **Theorem 9 (Switching-Type Spectrum Trend Theorem)** is refined:

$r(N)$ monotonically increases, $r(N) > 5$ (direction – reversal type dominant)

The regime switches of the nuclear shell are all biased toward the direction-reversal type (dominated by intercept mutation), with the switch intensity increasing monotonically as the nucleon number increases (Spearman $\rho = 0.9489$, $p = 0.0000$). Theorem 9 applies only to the 17 confirmed direction-reversal breakpoints (Bootstrap 95% CI lower bound > 5). The 3 light-nucleus-region candidate breakpoints at $N=14, 15$, and 29 , whose Bootstrap 95% CI lower bounds are < 5 , cannot exclude the rule-reset type.

5. Disappearance of Regime-Switch Signals under High-Precision Fitting of the Liquid-Drop Model

5.1 Liquid-Drop Model Residual Chow Signal Systematically Weaker than OLS Model

Table 8 reports the Chow F comparison between the liquid-drop model residuals and the OLS model residuals at 11 key breakpoints.

Table 8: Liquid-Drop Residual Chow vs. OLS Residual Chow (Comparison at 11 Breakpoints)

Breakpoint N	OLS Residual Chow F	Liquid-Drop Residual Chow F	Ratio (Liquid-Drop / OLS)
20	2104	239	0.11
28	1729	352	0.20
32	1546	354	0.23
50	877	422	0.48
65	438	390	0.89
82	236	345	1.46
83	217	343	1.58
104	55	238	4.33
126	17	182	10.71
129	12	174	14.50

In the light-nucleus region (N=20–32): the liquid-drop model residual Chow signal is 4–9 times weaker. In the heavy-nucleus region (N=104–129): the liquid-drop model residual Chow signal is instead stronger—possibly due to the decreased fitting accuracy of the liquid-drop model in the heavy-nucleus region.

The mean Chow F of the liquid-drop model residuals at the blind-detected breakpoints is 270.89, and the mean Chow F at random positions is 299.76 (± 152.80), with a Z-score of -0.19 and $p = 0.5749$ —the structural information of the regime switch is undetectable in the liquid-drop model residuals.

Ruling out the alternative explanation of "reduced test power due to low signal-to-noise ratio": By comparing the local signal-to-noise ratios (local window ± 8 neutrons) of the liquid-drop residuals and the OLS residuals at the blind-detected breakpoints, it is found that in the heavy-nucleus region ($N > 32$), the signal-to-noise ratio of the liquid-drop residuals is instead significantly higher than that of the OLS residuals (mean ratio 5.62), while the Chow F remains systematically lower (mean ratio 0.31)—ruling out the alternative explanation of "reduced test power due to low signal-to-noise ratio." The disappearance of the signal is entirely explained by the smooth fitting of the trends on both sides of the breakpoints by the liquid-drop model. In the light-nucleus region ($N \leq 32$), the signal-to-noise ratio of the liquid-drop residuals is slightly lower than that of the OLS residuals (mean ratio 0.74), and the reduction in the signal-to-noise ratio may partially contribute to the weakening of the liquid-drop Chow F in this region. Comprehensive judgment: the disappearance of the regime-switch signals in the liquid-drop model residuals is entirely explained by smooth fitting in the heavy-nucleus region and may mix the two effects of signal-to-noise ratio reduction and smooth fitting in the light-nucleus region.

5.2 Regime Jump Revealed by Hierarchical Regression

The full set of nuclides was divided into three groups by N —light nuclei ($N \leq 50$), medium nuclei ($50 < N \leq 100$), and heavy nuclei ($N > 100$)—and the same equation was fitted to each group separately. **Table 9** reports the results.

Table 9: Hierarchical Regression R^2 Comparison (Light / Medium / Heavy Nuclei)

Nuclear Region	Sample Size	R^2	N_over_Z Coefficient
Light nuclei	425	0.7049	−1466.21
Medium nuclei	635	0.9420	−998.0
Heavy nuclei	308	0.9911	−60.28

The N_over_Z coefficient jumps from −1466 in light nuclei to −60 in heavy nuclei—a jump by a factor of 24.4. The R^2 of the same equation differs enormously across different nuclear regions—the equation structure itself is different under different regimes.

5.3 Verification of the Predictive Ability of the Regime-Switch Model

Rolling-window cross-validation (window = 60, prediction step = 10, 639 windows in total) was deployed to compare the prediction RMSE of the global model and the segmented model (segmented according to the blind-detected breakpoints). **Table 10** reports the results.

Table 10: Segmented Model Prediction RMSE Improvement (by Nuclear Region, 639 Rolling Windows)

Nuclear Region	Number of Windows	Mean Improvement	Improvement Magnitude at Key Breakpoints
Light-nucleus region	162	1.7%	34% at $N=15$, 40% at $N=29$
Medium-nucleus region	259	1.0%	53% at $N=65$, 28% at $N=83$
Heavy-nucleus region	174	5.3%	48% at $N=104$, 66% at $N=118$
Superheavy-nucleus region	44	−2.8%	Sparse sample

The segmented model reduces the prediction RMSE by 30–66% near the blind-detected breakpoints. The global Wilcoxon test gives $p = 0.0036$ —the segmented model is significantly better than the global model. Regime switches are not overfitting noise but real data structures—when the blind-detected breakpoints are used to segment and predict future data, the prediction accuracy systematically improves. The improvement in the superheavy-nucleus region is negative (−2.8%) because the sample size in this region is too small (only 44 windows), and the parameter estimates of the segmented model are unstable.

5.4 Physical Implications

The high-precision fitting of the liquid-drop model renders the regime-switch signals undetectable in its residuals—this is a statistical fact. There are two possible physical mechanisms for this phenomenon: the liquid-drop model "absorbs" the regime-switch signals (encoding them in the fitting parameters) or "masks" the regime-switch signals (smoothing away the structural ruptures with a smooth function).

The shell-correction compensation traces in Section 6—the consistent avoidance of the shell-closure positions by three independent tools—support the "absorption" hypothesis: the success of the shell-correction term precisely demonstrates that it captures and absorbs the shell-closure signals, leaving the residuals free of these signals. Sections 5 and 6 together constitute a complete logical chain: Section 5 proves that the regime-switch signals disappear from the residuals of the high-precision model; Section 6 proves that this disappearance is because the shell-correction term "absorbs" these signals.

Regardless of whether the mechanism is "absorption" or "masking," high-precision models may systematically mask structural signals outside their framework—this is a methodological caution. While successfully describing physical phenomena, high-quality models may precisely erase the signals for discovering new physics. High R^2 does not equal being closer to the truth—fitting accuracy and structural discovery ability may be negatively correlated.

6. Compensation Traces of the Shell-Correction Term—Statistical Fingerprints Independently Detectable

The following three sub-evidences—the cross-regime rupture of the covariance (§6.3.1), the decay-mode interaction effects (§6.3.2), and the two-dimensional Chow test on doubly magic nuclei (§6.3.3)—describe different facets of the same physical phenomenon: the independent manifestations of the regime switch in the measurement-error structure, the decay-channel grouping, and the two-dimensional nuclide chart. They are not three independent discoveries but cross-validations of the same core discovery (the real existence of regime switches in nuclear physics) across three independent dimensions.

6.1 Reverse Significance of the Shell Order

The distributions of breakpoints at the shell-closure positions for the three blind-detection tools—PELT, Bai-Perron, and Bayesian segmentation—which are based on completely different mathematical principles, are shown in **Table 11**.

Table 11: Breakpoint Distributions of the Three Tools at Shell-Closure Positions

Shell-Closure N	Shell Name	PELT	Bai-Perron	Bayesian
2	1s1/2	0	0	0
8	1p1/2	0	0	0
20	1d3/2	0	0	0
28	1f7/2	0	1	1
40	2p1/2	0	0	0
50	1g9/2	0	0	0
82	1h11/2	0	1	1
126	1i13/2	0	0	0

The densities at the shell-closure positions for all three tools are lower than those at non-closure positions. The PELT density ratio is 0.100 vs. 0.263, the Bai-Perron ratio is 0.250 vs. 0.339, and the Bayesian ratio is 0.250 vs. 0.339. The Mann-Whitney U test gives $p = 0.0487$ —the difference is just significant at the 5% level. All three tools exhibit avoidance at the shell-closure positions—not because they cannot detect signals, but because their optimal breakpoints systematically deviate from the traditional magic numbers. It must be candidly acknowledged that $p = 0.0487$ is marginally significant; the consistent avoidance by the three tools is more critical evidence than the single p-value.

6.2 No-Shell-Correction Liquid-Drop Model Residual Test

A no-shell-correction liquid-drop model (containing only four terms: volume, surface, symmetry energy, and Coulomb, with parameters $a_v = 15.75$, $a_s = 17.8$, $a_a = 23.7$, $a_c = 0.71$ MeV) was constructed, and its residuals (experimental binding energy minus liquid-drop model prediction) were extracted. A Chow scan was performed on the residuals over the full N-axis.

The global Chow F of the no-shell-correction model residuals at the traditional magic numbers is extremely high—approximately 1820 at $N=20$, approximately 1880 at $N=28$, and approximately 1375 at $N=50$. However, there is almost no signal within the local window at the shell-closure positions—because the no-shell-correction model has not even fitted the basic trend well, and the local signal of the shell closure is submerged in the trend noise.

After the shell-corrected model (OLS 3-factor) successfully removes the overall trend, the local signal of the shell closure emerges at the correct physical position. What PELT detects are the residual structures in the shell-corrected model residuals—deformation transitions, pairing fluctuations, etc.—which are instead richer between the shell closures. This rules out the alternative explanation of "poor local data quality or statistical noise."

6.3 Multiple Independent Verifications

Cross-Regime Rupture of the Covariance: The same-side/cross-boundary ratio of the mass covariance is 22.31 (ratio of 87834 at N=14). Cross-regime rupture of reaction-energy errors: all 7 reaction-energy error columns (S2n_err_keV, S2p_err_keV, Sn_err_keV, Sp_err_keV, Qa_err_keV, QB_err_keV, Q2B_err_keV) exhibit systematic high/low ratio changes at all 9 regime boundaries. At N=126, the ratios for the 7 columns, 1.68–1.86, all have $p < 0.001$. The covariance chi-square is 6–12 billion: the regime switch transcends measurement errors.

Decay-Mode Interaction Effect Test: All four main decay modes satisfy the defining characteristics of a regime factor. **Table 12** reports the results.

Table 12: Decay-Mode Interaction Effect Test

Decay Mode	Interaction F	p-value	Judgment
decay_beta_minus	76.94	4.45×10^{-48}	Regime Factor
decay_beta_plus	36.24	4.96×10^{-23}	Regime Factor
decay_alpha	78.06	9.32×10^{-49}	Regime Factor
decay_stable	55.81	3.55×10^{-35}	Regime Factor

Stable nuclides have the lowest $R^2 = 0.37$ among all decay-channel groups—they are the "boundary sentinels" of the nuclear-physics version.

Systematic Two-Dimensional Chow Test on Doubly Magic Nuclei: All five doubly magic nuclei with sufficient data pass the two-dimensional Chow test. **Table 13** reports the results.

Table 13: Systematic Two-Dimensional Chow Test on Doubly Magic Nuclei (5 Doubly Magic Nuclei)

Doubly Magic Nucleus	(N, Z)	Chow F	Minimum Cell Size
^{40}Ca	(20, 20)	508.88	15
^{48}Ca	(28, 20)	420.71	55
^{56}Ni	(28, 28)	406.33	10
^{132}Sn	(82, 50)	230.31	27
^{208}Pb	(126, 82)	162.25	45

All five doubly magic nuclei pass the two-dimensional Chow test, with F decreasing monotonically from light to heavy.

6.4 Physical Implications: The Success of the Shell Correction = Compensation for the Insufficiency of Independent Particle Motion

The shell-correction method (Strutinsky method) was introduced to remedy the discrepancy between independent particle motion (the shell model) and experiment—it is a phenomenological compensation for

many-body correlation effects. If independent particle motion is the fundamental organizing principle, why does it require such a massive external correction?

"Reverse significance" is not evidence that shell closures do not exist but evidence that the shell-closure signals have been precisely compensated in the high-quality model. This is direct evidence that "independent particle motion requires external compensation"—the shell model is an approximation, not a first principle. The core value of this discovery lies in the fact that it reveals statistical fingerprints left by high-quality theoretical models in the data, fingerprints that can be independently detected by the operating system of the Factor Hierarchy Law.

7. Cascade Skeleton

7.1 Cascade Alignment of the Confirmed Breakpoints

A systematic comparison of the 5 confirmed breakpoints [23, 89, 108, 129, 145] with the traditional magic numbers is performed. **Table 14** reports the cascade alignment results.

Table 14: Cascade Alignment of Traditional Magic Numbers and Confirmed Breakpoints

Traditional Magic Number	Pre Confirmed Breakpoint	Post Confirmed Breakpoint	Cascade Characteristic
N=20	None	23	Only post-clamped (lightest end of the light-nucleus region)
N=28	23	89	Shares N=23 with N=20 as the pre confirmed breakpoint
N=50	23	89	Shares the [23, 89] clamping interval with N=28
N=82	23	89	Shares the [23, 89] clamping interval with N=28 and N=50
N=126	108	129	Independent clamping interval

Every traditional magic number is clamped by at least one confirmed breakpoint, without exception. N=20 and N=28 share N=23 as a cascade node—the post confirmed breakpoint of N=20 (=23) is exactly the pre confirmed breakpoint of N=28 (=23), with a cascade gap of zero.

7.2 Statistical Characteristics of N=89 Simultaneously Clamping Three Traditional Magic Numbers

N=89 simultaneously clamps the three traditional magic numbers N=28, N=50, and N=82. Quantitative analysis shows that the distances from the three magic numbers to N=89 differ enormously (61 neutrons for N=28, 39 neutrons for N=50, and 7 neutrons for N=82), yet they share the same statistical boundary. The local

Chow signal attenuates with distance— $F = 0.21$ at $N=28$ (almost silent), $F = 1.17$ at $N=50$, and $F = 1.51$ at $N=82$.

This attenuation pattern is inconsistent with the physical picture of the shell model, in which "each shell closure is independently determined by different single-particle energy gaps"—if the three shell closures were completely independent, they should not share the same statistical boundary, nor should they attenuate with distance. The Factor Hierarchy Law provides a natural explanation: $N=89$ is the macroscopic regime boundary between the medium-heavy and heavy-nucleus regions, and the shell closures of $N=28$, $N=50$, and $N=82$ share this statistical boundary in the binding-energy equation.

It must be candidly acknowledged that the possibility that the one-to-many clamping arises from the sparsity of the confirmed breakpoints cannot currently be excluded. Five confirmed breakpoints cover seven traditional magic numbers; $N=89$ is the only confirmed breakpoint falling within the $N=50$ – 82 interval and naturally becomes the shared boundary for $N=28$, $N=50$, and $N=82$. If more confirmed breakpoints are discovered in the $N=28$ – 82 interval in the future, and they each independently align with different traditional magic numbers, this would support the exact cascade hypothesis; if the newly added breakpoints still exhibit a one-to-many clamping pattern, this would support the current shared-boundary interpretation. For now, this observation is positioned as a statistical feature awaiting clarification and is not accorded the status of a "discovery."

7.3 Theorem 8: Confirmed-Breakpoint Cascade Skeleton Theorem

Based on the above findings, **Theorem 8 (Confirmed-Breakpoint Cascade Skeleton Theorem)** is refined:

$$T_2(20) = T_{\text{precursor}}(28) = 23$$

The 5 confirmed breakpoints [23, 89, 108, 129, 145] constitute a cascade skeleton covering all traditional magic numbers. The exact cascade between $N=20$ and $N=28$ holds— $T_2(20)=23$ is exactly equal to $T_{\text{precursor}}(28)=23$, with a cascade gap of zero. $N=89$ is simultaneously the fluctuation-texture-change point of the shell closures of the three traditional magic numbers $N=28$, $N=50$, and $N=82$ —the cascade degenerates from "precise transmission between adjacent magic numbers" to "one confirmed breakpoint simultaneously serving as the statistical boundary for multiple magic numbers," reflecting the fundamental difference between the continuity of regime factors in nuclear physics and the discrete symmetries in particle physics. Cascade precision is a measure of the "purity" of the regime factor.

7.4 Physical Implications of the Confirmed-Breakpoint Gaps

The gap $23 \rightarrow 89 = 66$ covers the range of 55 neutrons from $N=28$ to $N=82$ —there is no other breakpoint passing cross-validation between the exit of the light-nucleus region and the entrance of the medium-nucleus

region. The gaps $89 \rightarrow 108 = 19$ and $108 \rightarrow 129 = 21$ exactly clamp the traditional magic number $N=126$. The gap $129 \rightarrow 145 = 16$ enters the superheavy-nucleus region.

8. Independent Verification by HMM and Kernel Change-Point

HMM and kernel change-point are not independent parallel analyses—they independently verify two core characteristics of the confirmed-breakpoint network from the perspectives of hidden-state decoding and nonlinear covariance detection, respectively: the reasonableness of the number of shell regions and the authenticity of the precursor signal in the light-nucleus region.

8.1 HMM 2–8 Hidden-State BIC Selection

Table 15: HMM 2–8 Hidden-State BIC Selection

Number of Hidden States	BIC	Δ BIC
2	6250.3	—
3	6258.4	+8.1
4	2948.5	−3309.9
5	2951.7	+3.2
6	2747.0	−204.7
7	2322.1	−424.9
8	1646.9	−675.2

The BIC decreases monotonically from 2 states to 8 states, without a minimum turning point appearing at 8 states. The BIC drops sharply by 3310 when transitioning from 3 states to 4 states—the binding-energy data require at least 4 hidden states to basically describe its regime structure, rather than the 2 states (liquid-drop + shell correction) assumed by traditional nuclear physics. The physical picture corresponding to the 4 states is: the light-nucleus region, the medium-nucleus region, the heavy-nucleus region, and the superheavy-nucleus region—four independent regimes. The sharp drop to 4 states in the HMM is highly consistent with the 4 shell regions partitioned by the 5 confirmed breakpoints—two completely independent methods (regression structural-breakpoint detection and probabilistic graphical model state decoding) converge on the same answer for the number of shell regions.

It must be candidly acknowledged that the HMM BIC has not shown a minimum turning point at 8 states, meaning that the optimal number of hidden states may be greater than 8. This is not fully consistent with the 4 shell regions partitioned by the confirmed breakpoints—the 4 shell regions are a conservative estimate based on the 5 confirmed breakpoints, and the HMM may be capturing finer regime structures. The difference in the number of shell regions between the two methods does not constitute a contradiction—they are two independent

estimates of the same regime-switch network at different resolutions. The consistency between the sharp BIC drop of 3310 at the 3→4 state transition and the 4 shell regions of the confirmed breakpoints is the lowest estimate on which the two methods converge for the number of core shell regions.

8.2 Comparison of Breakpoints Detected by Three Kernel Functions of the Kernel Change-Point

Table 16: Comparison of Breakpoints Detected by Three Kernel Functions of the Kernel Change-Point

Kernel Function	Breakpoints N	Count	Distribution Characteristic
linear	[6, 11, 18, 30, 69, 88, 121, 149]	8	Uniformly distributed, from N=6 to N=149
rbf	[21, 55, 69, 88, 108, 128, 141, 156]	8	Concentrated in the medium-heavy nucleus region
cosine	[4, 6, 8, 9, 10, 12, 13, 14]	8	All concentrated at N < 15

All 8 breakpoints of the cosine kernel are concentrated at $N < 15$ —constituting an independent cross-validation with the inv-family activation operators locking onto $N=14$ in Tang (2026v). The Tool-Dependence Theorem receives its most extreme verification in nuclear physics—the differences in the breakpoint positions between different kernel functions are so large that the cosine kernel concentrates entirely at $N < 15$ while the linear kernel distributes uniformly up to $N=149$.

9. Discussion

9.1 Unification of "Magic-Number Shift" and "Boundary Alignment"—Two Sides of the Same Transition-Zone Structure

The traditional shell model regards a shell closure as a point—a magic number. The Factor Hierarchy Law finds that the shell closure unfolds, in a finite nuclear system, into a regime-switch zone with a well-defined internal structure and geometric boundaries. The "shift" (Tang, 2026v) measures the distance between two points within the transition zone (the quantum-mechanical starting point and the statistical endpoint)—the magnitude of the shift is associated with the physical type of the shell closure. The "alignment" (this paper) measures the clustering of the boundaries between the transition zones according to the traditional magic-number shell regions—the precision of the alignment is quantitatively characterized by the zero deviation of the Transition-Zone Width Theorem.

The two are not contradictory findings but two complementary perspectives on the same physical picture: the shift reveals that the shell closure is a "zone" rather than a "point," and the alignment reveals that the

boundaries of the "zones" cluster exactly according to the traditional shell regions. This unified understanding is the empirical foundation for the subsequent demotion of the shell model (Tang, 2026x).

9.2 Comparison with the Precursor Cascade in Particle Physics

In particle physics (Tang, 2026o): exact cascades hold for both cascade pairs—strange $T_2 \approx$ charm T_1 (deviation 0.0052 GeV), and charm $T_2 =$ bottom precursor (deviation 0.0000 GeV). In nuclear physics: the exact cascade holds only for $N=20 \rightarrow 28$ ($T_2=23 =$ precursor=23), while the medium-heavy nucleus region degenerates into one confirmed breakpoint ($N=89$) simultaneously serving as the statistical boundary for three traditional magic numbers. The physical root: in particle physics, color charge is a strict discrete symmetry, and the cascade precision can reach 10^{-4} GeV; in nuclear physics, the neutron number is an approximate regime factor for multiple interactions, and the cascade precision is 1 neutron. Cascade precision is a measure of the "purity" of the regime factor.

9.3 Summary of New Discovery Assets

Table 17: Fifteen New Discoveries Completely Unknown to Traditional Nuclear Physics

No.	Discovery	Core Content	Positioning	Argument Strength	Depends on Candidate Breakpoints?
1	Transition-Zone Width Theorem	Internal self-consistency of the confirmed-breakpoint network—4 shell regions with zero deviation	Theorem	★★★★	No
2	Shell-model coverage only 15%	Only 3 of the 20 blind-detected breakpoints fall within ± 2 of traditional magic numbers	Exploratory Assessment	★★	Yes
3	Odd-even effect magic-number jump	Halving at N=20, doubling at N=28, reversing at N=82	Genuine New Discovery	★★	No
4	N=Z line as a regime boundary	Optimal breakpoints of 111 isotope chains in the light-nucleus region tightly follow the N=Z line	Genuine New Discovery	★★	No
5	Cascade skeleton	5 confirmed breakpoints constitute a cascade skeleton covering all traditional magic numbers	Theorem	★★	No
6	Signal disappearance in liquid-drop residuals	$R^2=0.96$ but residual Chow signal 4–9 times weaker	Core Evidence	★	No
7	Shell-correction compensation traces	Three tools consistently avoid, covariance rupture ratio 22.31	Supplementary Evidence	★	No
8	Switching-type monotonic increase	17 confirmed breakpoints of direction-reversal type, Spearman $\rho=0.95$	New Empirical Regularity	★★	Yes
9	Decay mode as a regime factor	All four decay-mode interaction effects $p < 10^{-23}$	New-Form Regime Factor	★★	No

No.	Discovery	Core Content	Positioning	Argument Strength	Depends on Candidate Breakpoints?
10	Stable-nuclide boundary sentinel	Lowest $R^2=0.37$ but extremely significant interaction effect	Known Phenomenon Revealed by Statistical Audit	★	No
11	Doubly magic nuclei all pass 2D Chow	All five doubly magic nuclei pass, F decreasing monotonically from light to heavy	New Empirical Regularity	★★	No
12	Segmented model predictive advantage	RMSE drops 30–66% near blind-detected breakpoints	Supplementary Evidence	★★	Yes
13	HMM sharp drop at 4 states	BIC drops 3310 at 3→4 state transition, consistent with 4 shell regions	Independent Cross-Validation	★★	No
14	Kernel CP cosine kernel concentration	All 8 breakpoints concentrated at $N < 15$	Extreme Verification of Tool Dependence	★	No
15	Traditional magic-number gap monotonic increase	$N=20 \rightarrow 28$ gap 0, $28 \rightarrow 50$ gap 21, etc.	New Empirical Regularity	★★	No

Argument Strength Labels: ★★★ = Theorem (passed rigorous statistical testing), ★★ = Empirical Regularity or Independent Cross-Validation, ★ = Supplementary Evidence or Known Phenomenon Revealed by Statistical Audit. The label "Depends on Candidate Breakpoints" indicates that the discovery incorporates information from the candidate breakpoints, and the argument strength is correspondingly reduced.

9.4 Methodological Contributions and Limitations

This paper systematically introduces statistical significance testing into the screening process of multi-tool aggregated breakpoints and constructs a methodological template of multi-level defense tests—when statistical signals face challenges of "test failure" or "over-pruning," how to distinguish real structures from artifacts through candid acknowledgment of argument boundaries.

Candid Disclosures: (1) The permutation test exhibits statistical conservatism under the strong inherent structure of the nuclear binding-energy data—more refined testing methods may restore the significance of

some single-tool breakpoints; (2) The 5 confirmed breakpoints constitute a conservative but solid cascade skeleton; the fine structure of the cascade network requires larger samples and finer tools for verification; (3) The HMM BIC has not shown a minimum turning point at 8 states, and the optimal number of hidden states may be greater than 8, which is not fully consistent with the 4 shell regions partitioned by the confirmed breakpoints—the two methods are two independent estimates of the same regime-switch network at different resolutions; (4) The reverse significance of the shell order has been independently verified by three tools, but whether the avoidance mechanisms of the various tools at the shell-closure positions are the same requires further investigation; (5) The 15 candidate breakpoints provide testable candidate targets for subsequent research; (6) The random-breakpoint-set control test is positioned as a sensitivity analysis; the self-consistency of Theorem 7 relies primarily on the internal consistency of the network, and the alignment accuracy with externally defined shell-region boundaries based on traditional magic numbers is not significantly different from the random expectation ($p = 0.4197$); (7) The 6-shell-region version is positioned as a method demonstration and does not constitute an independent verification of Theorem 7; (8) Distinguishing the causal direction of the odd-even effect requires time-series data or independent experiments; (9) The MWU test $p = 0.0487$ is marginally significant; the consistent avoidance by the three tools is more critical evidence than the single p -value.

10. Conclusion

Based on the regime factors and execution factors locked by Tang (2026v), this paper deployed four newly added structural breakpoint tools and permutation test screening, screening out 5 confirmed breakpoints [23, 89, 108, 129, 145] from the 20 multi-tool aggregated blind-detected breakpoints.

The core findings are distilled into five tiers of empirical discovery and three quantitative theorems (Theorems 7–9):

Tier 1—Transition-Zone Width Theorem (Theorem 7): Establishes the internal self-consistency of the confirmed-breakpoint network—the confirmed breakpoints fall exactly on the shell-region boundaries defined by one another, organized according to precise topological regularities. Random noise cannot automatically arrange itself into a structure with zero deviation (1000 random breakpoint sets never reproduced 7 exact hits, $p < 0.001$). The self-consistency of Theorem 7 relies primarily on the internal consistency of the network; the alignment accuracy with externally defined shell-region boundaries based on traditional magic numbers is not significantly different from the random expectation.

Tier 2—New structures unexplained by the shell model: Shell-model coverage is only 15% (exploratory assessment), testing whether "magic numbers are not the optimal description of the statistical structure" rather than whether "the shell model is wrong"; the odd-even effect undergoes systematic jumps at magic numbers;

optimal breakpoints of 111 isotope chains in the light-nucleus region tightly follow the $N=Z$ line, identifying the $N=Z$ line itself as a regime boundary; 17 of the 20 breakpoints are confirmed as direction-reversal type, with the intercept/slope ratio satisfying a monotonic increasing trend (Theorem 9 applies only to these 17 confirmed breakpoints).

Tier 3—Disappearance of regime-switch signals under high-precision fitting of the liquid-drop model: The liquid-drop model achieves $R^2 = 0.96$, but its residual Chow signal is 4–9 times weaker in the light-nucleus region. In the heavy-nucleus region, the signal-to-noise ratio of the liquid-drop residuals is significantly higher than that of the OLS residuals (mean ratio 5.62), while the Chow F remains systematically lower (mean ratio 0.31), ruling out the alternative explanation of "reduced test power due to low signal-to-noise ratio." The shell-correction compensation traces in Section 6 support the "absorption" hypothesis. High-precision models may systematically mask structural signals outside their framework—this is a methodological caution.

Tier 4—Compensation traces of the shell-correction term: Three independent tools consistently avoid the shell-closure positions; the no-shell-correction model test rules out alternative explanations; the covariance rupture ratio is 22.31; reaction-energy errors show systematic jumps across all 7 columns $\times$ 9 boundaries; all four decay-mode interaction effects have $p < 10^{-23}$.

Tier 5—Cascade skeleton (Theorem 8): $N=20$ and $N=28$ share $N=23$ as a cascade node, establishing exact cascade alignment; $N=89$ simultaneously clamps $N=28$, $N=50$, and $N=82$ —three traditional magic numbers—with the local Chow signal attenuating with distance. The cascade degenerates into a shared boundary, reflecting the fundamental difference between the continuity of regime factors in nuclear physics and the discrete symmetries in particle physics. The possibility that the one-to-many clamping arises from the sparsity of the confirmed breakpoints cannot currently be excluded; this observation is positioned as a statistical feature awaiting clarification.

This paper provides the last three key pieces of evidence (Evidence 3–5) promised by Tang (2026v) that "independent particle motion is not the fundamental organizing principle": shell-model coverage of only 15%, disappearance of regime-switch signals under high-precision fitting of the liquid-drop model, and the shell-correction term requiring external compensation. Together with Tang (2026v), this paper completes the systematic audit of the completeness of the shell model.

This paper reports a total of fifteen new discoveries completely unknown to traditional nuclear physics (Table 17) and deploys a methodological template of multi-level defense tests. The subsequent paper (Tang, 2026x) will, based on all the findings of this paper and Tang (2026v), argue that independent particle motion is not the fundamental organizing principle of the nuclear shell and formally complete the demotion of the shell model.

Data Availability Statement

All data in this study come from the international Atomic Mass Evaluation AME2020 and the nuclear properties database NUBASE2020, obtained through the official channel of the IAEA Atomic Mass Data Center (AMDC). The data are completely public, and any researcher can independently reproduce all results. The analysis code is available upon request.

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